Generative Neural Networks for the Sciences

Evaluating Synthetic Data Augmentation for Low-Data Breast Lesion Segmentation in Ultrasound Images

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Code Repository: https://github.com/RoshCardoza/GNN_FInal_Project.git

Lecture: Generative Neural Networks for the Sciences

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1 Abstract

Main author: Bhavana Krishna

Medical image segmentation is a central task in computer-aided diagnosis because it allows clinically relevant structures to be localized and measured directly from imaging data. In practice, however, accurate segmentation models usually require large annotated datasets, while medical annotation is expensive and time-consuming. This project investigates whether synthetic data augmentation can improve breast lesion segmentation in ultrasound images when only a limited amount of labeled training data is available.

We use the BUSI breast ultrasound dataset and compare three training settings: a real-only baseline, classical geometric augmentation, and synthetic augmentation based on lesion copy-paste. All experiments are conducted in a controlled low-data setup using only 5%, 10%, and 25% of the training split. A U-Net segmentation model is trained and evaluated using Dice score and Intersection over Union (IoU). In addition to quantitative evaluation, qualitative prediction examples are examined to understand the strengths and failure cases of each approach.

The project is motivated by the observation that synthetic data are often proposed as a solution to annotation scarcity, but their usefulness is not guaranteed. A synthetic image may appear plausible while still containing unrealistic boundaries, texture inconsistencies, or structural artifacts that harm downstream learning. For this reason, the project focuses not only on producing synthetic images, but on determining whether these images improve the segmentation task itself.

An exploratory GAN-based pipeline was also implemented, but the final experiments concentrate on lesion copy-paste augmentation because it proved more stable and practical for the BUSI dataset and the course time constraints. Our results show the real-only baseline improved steadily from the 5% to the 25% data regime, confirming that additional real data remain highly valuable. Classical augmentation produced limited and inconsistent gains, whereas synthetic augmentation showed a more context-dependent effect, appearing more beneficial when a moderate Medical image segmentation is a central task in computer-aided diagnosis because it allows clinically relevant structures to be localized and measured directly from imaging data. In practice, however, accurate segmentation models usually require large annotated datasets, while medical annotation is expensive and time-consuming. This project investigates whether synthetic data augmentation can improve breast lesion segmentation in ultrasound images when only a limited amount of labeled training data is available.

We use the BUSI breast ultrasound dataset and compare three training settings: a real-only baseline, classical geometric augmentation, and synthetic augmentation based on lesion copy-paste. All experiments are conducted in a controlled low-data setup using only 5%, 10%, and 25% of the training split. A U-Net segmentation model is trained and evaluated using Dice score and Intersection over Union (IoU). In addition to quantitative evaluation, qualitative prediction examples are examined to understand the strengths and failure cases of each approach.

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An exploratory GAN-based pipeline was also implemented, but the final experiments concentrate on lesion copy-paste augmentation because it proved more stable and practical for the BUSI dataset and the course time constraints. Our results show the real-only baseline improved steadily from the 5% to the 25% data regime, confirming that additional real data remain highly valuable. Classical augmentation produced limited and inconsistent gains, whereas synthetic augmentation showed a more context-dependent effect, appearing more beneficial when a moderate amount of real training data was already available. This suggests that synthetic augmentation is not a universal replacement for real data, but can serve as a useful complement under suitable conditions. Overall, the project provides a systematic evaluation of synthetic augmentation for low-data breast ultrasound lesion segmentation and suggests that the usefulness of generated data should be judged primarily by downstream task performance rather than by visual plausibility alone.

In addition, this study highlights an important practical issue in medical image analysis: methods that appear promising in theory do not always translate into measurable gains in real experimental settings. By evaluating each augmentation strategy under the same controlled data regimes, the project aims to provide a fair comparison of how different forms of data expansion influence segmentation quality. The work also emphasizes reproducibility and task-oriented validation by keeping the segmentation architecture fixed and varying only the training data strategy. This makes it easier to isolate the effect of synthetic augmentation itself. From a clinical AI perspective, the findings are relevant because robust low-data learning remains essential in domains where expert annotations are scarce. Therefore, the contribution of this project is not only an experimental comparison, but also a practical assessment of when synthetic data may be helpful, when it may offer little benefit, and why careful downstream evaluation is necessary before adopting augmentation methods in medical imaging workflows.amount of real training data was already available. This suggests that synthetic augmentation is not a universal replacement for real data, but can serve as a useful complement under suitable conditions. Overall, the project provides a systematic evaluation of synthetic augmentation for low-data breast ultrasound lesion segmentation and suggests that the usefulness of generated data should be judged primarily by downstream task performance rather than by visual plausibility alone.

2 Introduction

Main author: Roshal Cardoza

Medical image analysis has become one of the most important application areas of machine learning. Neural networks are now widely used for classification, detection, segmentation, reconstruction, registration, and image enhancement in medicine. Among these tasks, **segmentation**

is especially important because it produces detailed spatial information rather than a single image-level label. Instead of merely deciding whether an abnormality is present, a segmentation system attempts to show exactly where it is, how large it is, and what shape it has. This makes segmentation useful in many clinical scenarios, including lesion measurement, disease monitoring, treatment planning, and image-guided intervention. In many diagnostic workflows, accurate delineation of anatomical or pathological structures is essential not only for visual interpretation but also for quantitative analysis. For this reason, segmentation is often a foundational step that supports later decision-making and downstream clinical applications.

This project focuses on **breast lesion segmentation in ultrasound images**. Ultrasound is widely used in clinical practice because it is non-invasive, inexpensive, and easy to repeat. It is especially useful as a complementary examination when suspicious tissue has already been identified by palpation or by another imaging modality such as mammography. Ultrasound can help distinguish cystic from solid lesions and can support image-guided diagnostic procedures. However, from the perspective of automated analysis, ultrasound is a difficult modality. Images contain speckle noise, weak contrast, irregular intensity patterns, and shadowing artifacts. Lesions can vary considerably across patients and acquisition conditions, and normal tissue structures may resemble lesions. These characteristics make the boundary between lesion and surrounding tissue difficult to define, even for human experts, and they increase the difficulty of learning robust segmentation models. As a result, breast ultrasound segmentation represents a realistic and clinically meaningful test case for evaluating low-data learning strategies.

Deep learning models such as U-Net have become standard tools for medical image segmentation because they can learn multi-scale features while preserving local spatial information. Yet these models typically need a substantial amount of annotated training data. This creates a practical problem in the medical domain. Although clinical imaging produces large amounts of raw data, pixel-wise annotation is expensive because it usually requires expert input. In segmentation, the annotation burden is even higher than in image-level classification, since the annotator must define the lesion boundary rather than simply providing a single global label. In addition, inter-observer variability can affect annotation consistency, especially in modalities such as ultrasound where lesion margins are often ambiguous. Therefore, the challenge is not only a lack of data quantity, but also the difficulty of obtaining annotations that are both precise and reliable.

This makes **low-data learning** an important and practically relevant problem. A low-data setting is not merely a small dataset in an absolute sense. Rather, it is a situation in which the amount of labeled data available for training is not sufficient for the model to learn robust and generalizable features in a stable way. In segmentation, this challenge is amplified because the network must learn both where lesions occur and how their boundaries are shaped. If the training set is too small, the model may overfit to specific cases, fail to generalize to new lesion appearances, or default too often to trivial predictions such as empty masks. Low-data conditions are especially relevant in medicine because many clinically important problems involve rare conditions, institution-specific data distributions, or strict annotation budgets. Consequently, methods that can improve performance under limited supervision are of high practical value.

A common strategy for addressing limited data is **data augmentation**. Classical augmentation creates additional training examples by applying transformations to existing images and masks. Examples include horizontal flips, vertical flips, rotations, and mild geometric perturbations. These techniques are simple, widely used, and often effective because they increase variability in the training set without changing the underlying semantic meaning of the sample. However, classical augmentation can only produce transformed versions of existing images. It cannot create genuinely new lesion appearances or new image-mask pairs. In other words, it improves robustness to certain variations but does not fundamentally expand the diversity of the lesion patterns seen during training. This limitation becomes especially important when the available real samples are very few or do not adequately cover the range of lesion morphology present in practice.

This motivates **synthetic data augmentation**. Instead of merely transforming existing examples, synthetic augmentation aims to create new training data. In recent years, this idea has become especially prominent in the context of generative neural networks. Models such as GANs, VAEs, and diffusion models have been proposed for medical image generation, with the hope that synthetic data can improve performance when labeled real data are scarce. At the same time, the usefulness of synthetic data is not guaranteed. A generated image may appear plausible while still containing artifacts, unrealistic lesion boundaries, or incorrect texture statistics that make it harmful for downstream learning. Therefore, synthetic data should not be evaluated solely by their visual quality. Their real value lies in whether they improve the performance of the final task model. This distinction is particularly important in medical imaging, where visually appealing synthetic images may still fail to preserve subtle clinical characteristics that influence segmentation quality.

The central research question of this project is therefore:

> **Can synthetic data augmentation improve breast lesion segmentation performance in low-data ultrasound settings?**

To make this question more concrete, we consider three sub-questions:

1. Does synthetic augmentation improve segmentation performance compared with a real-only baseline?
2. How does synthetic augmentation compare with standard classical augmentation?
3. Under which low-data regimes does synthetic augmentation help, and under which regimes does it fail to provide clear benefit?

These questions are scientifically relevant because synthetic data augmentation is often treated as promising by default, but in practice its value depends strongly on the quality of the synthetic samples, the size and diversity of the real dataset, and the nature of the downstream task. In a medical context, answering this question carefully matters because clinical applications demand robustness, interpretability, and reproducibility. A small performance gain achieved

through unstable or unrealistic synthetic samples may not be meaningful in practice. Moreover, evaluating augmentation methods under controlled experimental settings is necessary to separate genuine improvement from random variation or overfitting. A careful comparison can therefore contribute not only to model development but also to a more realistic understanding of what synthetic data can and cannot provide.

The objective of this study is therefore not only to build a segmentation model, but to evaluate augmentation strategies under controlled conditions. We use the BUSI breast ultrasound dataset, construct a stratified train/validation/test split, and then simulate low-data regimes by training on only 5%, 10%, and 25% of the training split. We compare three settings: a real-only baseline, classical augmentation, and synthetic augmentation based on lesion copy-paste. In addition, an exploratory GAN-based notebook was implemented, but the final experiments focus on the more stable copy-paste strategy. The segmentation backbone is U-Net, which provides a strong and widely accepted baseline for biomedical image segmentation. Model performance is evaluated using Dice score and Intersection over Union, and qualitative prediction examples are also examined to better understand error patterns, strengths, and failure cases. This combination of quantitative and qualitative analysis helps ensure that the evaluation is not reduced to a single metric alone.

Several limitations should be stated explicitly. First, the project uses only one public dataset, BUSI, so the conclusions cannot automatically be generalized to all ultrasound segmentation tasks. Second, the main segmentation architecture is U-Net, which is a strong but not unique choice. Third, the final synthetic augmentation method is based on lesion copy-paste rather than a fully learned generative image model. Finally, the work was carried out within the time and resource constraints of a course project. Nevertheless, these constraints do not weaken the central contribution of the study. Instead, they motivate a focused and realistic design that aims to answer the research questions in a controlled and reproducible way. Overall, the project contributes a systematic examination of synthetic augmentation for breast ultrasound segmentation in limited-data conditions and highlights an important lesson for medical AI research: the value of synthetic data must be judged primarily by downstream task performance, not by visual plausibility alone. In this sense, the study offers both a practical experimental comparison and a broader methodological perspective on how augmentation strategies should be assessed in applied medical machine learning.

3 Background

3.1 Breast ultrasound lesion segmentation

Main author: Bhavana Krishna

Breast ultrasound is commonly used when suspicious tissue needs further examination. It is safe, relatively affordable, and does not expose patients to ionizing radiation. Compared with more structured modalities such as CT or MRI, however, ultrasound images are often difficult to analyze automatically. They contain speckle noise, low contrast, irregular intensity

patterns, and shadowing artifacts. Lesions vary in size, shape, brightness, and texture, and their boundaries may be weak or only partially visible. In addition, normal anatomical structures sometimes resemble pathological patterns, which increases the difficulty of distinguishing lesion from background.

From a machine learning perspective, lesion segmentation in ultrasound is a **dense prediction** problem. The network must decide for every pixel whether it belongs to the lesion or to the background. This makes the task more demanding than image-level classification. The model must not only detect that a lesion is present, but also localize it precisely. Even relatively small boundary errors can strongly affect overlap-based metrics such as Dice and IoU.

The BUSI dataset used in this project includes benign, malignant, and normal images. This is important because the normal images contain no lesion at all. A robust system must therefore handle two kinds of situations: images in which a lesion must be segmented accurately and images in which the correct output is an empty mask. This adds realism to the task but also introduces class imbalance and increases the chance that a model may fall back on overly frequent empty-mask predictions.

3.2 Why low-data segmentation is difficult

Main author: Bhavana Krishna

Low-data segmentation is difficult for several reasons. First, segmentation models solve a more complex task than image-level classifiers. A classifier only needs to decide whether an image belongs to one category or another. A segmentation model must label every pixel. This means it must learn both lesion appearance and lesion geometry, including detailed boundary structure.

Second, segmentation is usually affected by strong class imbalance. In many medical images, the lesion occupies only a small fraction of the total image area while the background covers most pixels. A model can therefore reach seemingly reasonable pixel-wise behavior while still failing to segment the lesion properly. This is why overlap-based losses and metrics are necessary.

Third, small datasets often fail to capture the full diversity of lesion appearance. Lesions may differ in echogenicity, size, shape, texture, edge sharpness, and surrounding tissue context. If only a limited number of these variations appear in the training set, the model may learn narrow patterns that do not transfer well to new cases.

Finally, segmentation is highly sensitive to annotation quality. Since the ground-truth mask defines the exact region the network is meant to learn, inconsistencies or ambiguities in annotation can directly affect performance. This makes low-data segmentation a meaningful research setting because it combines limited quantity, limited diversity, class imbalance, and sensitivity to label quality.

3.3 Dataset

Main author: Bhavana Krishna

This project uses the **BUSI (Breast Ultrasound Images Dataset)**, a public dataset of breast ultrasound images with three categories: benign, malignant, and normal. The commonly cited version contains 780 images from 600 female patients, including 437 benign images, 210 malignant images, and 133 normal images. The lesion-containing images are accompanied by annotation masks, while the normal images naturally do not contain lesion masks.

The BUSI dataset is well suited to this project for several reasons. First, it is public and widely used, which makes the work reproducible and easier to compare with related studies. Second, it contains both lesion masks and normal images, which supports a realistic segmentation pipeline. Third, its size is small enough to make low-data experiments meaningful rather than artificial.

In our workflow, the dataset is not treated as a single pool from which images are randomly drawn for training. Instead, we first create a **stratified train/validation/test split**, and only then generate smaller low-data subsets from the training split. This is important because it preserves a proper evaluation protocol. The validation and test data remain fixed and unseen during training, while the training data can be systematically reduced to simulate annotation scarcity.

Another important detail is how the normal images are handled. Since they contain no lesions, their masks are represented as empty binary masks. This makes the segmentation problem more realistic but also introduces a special case for evaluation metrics. If both the prediction and the target mask are empty, the result should not automatically be treated the same as a failed lesion localization case. Our implementation addresses this issue explicitly in the metric computation.

3.4 U-Net for medical image segmentation

Main author: Roshal Cardoza

U-Net is one of the most influential architectures in medical image segmentation. Originally designed for biomedical image analysis, it has become a standard baseline because of its relatively simple structure and strong empirical performance. The basic idea behind U-Net is to combine an encoder-decoder architecture with skip connections. The encoder progressively reduces the spatial resolution of the image while extracting increasingly abstract features. The decoder then upsamples these features back to the original image resolution and produces a dense pixel-wise prediction.

The skip connections are important because they preserve fine-grained spatial information that might otherwise be lost during downsampling. Since segmentation depends heavily on accurate localization, these connections make it easier to reconstruct object boundaries in the final mask.

For this project, U-Net is an appropriate choice for several reasons. First, it is a strong and widely accepted baseline. Second, it performs well even when the amount of training data is relatively limited. Third, it is computationally manageable within the available course resources. Since the main purpose of the study is to compare augmentation strategies rather than segmentation architectures, using U-Net helps keep the experimental focus clear.

3.5 Loss functions and evaluation metrics

Main author: Bhavana Krishna

Segmentation tasks are often strongly affected by class imbalance because the background region may occupy most of the image while the lesion occupies only a small part. A loss function that looks only at pixel-wise correctness can therefore be misleading, since a model may appear to perform well by predicting background too often. For this reason, overlap-based losses and evaluation metrics are especially important.

In this project, the segmentation model is trained with a combination of **binary cross-entropy (BCE) loss** and **Dice loss**. BCE loss compares the predicted logits with the binary target mask at the pixel level. It provides stable supervision during optimization. Dice loss, by contrast, emphasizes overlap between predicted and true foreground regions, making it more sensitive to whether the lesion is segmented accurately. Using both together provides a practical balance between stable training and task-relevant overlap quality.

The Dice score is defined as

$$\mathrm{Dice}(A, B) = \frac{2|A \cap B|}{|A| + |B|},$$

where A is the predicted lesion mask and B is the ground-truth mask.

The Intersection over Union is defined as

$$\mathrm{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

Dice and IoU are closely related, but IoU is stricter because the denominator includes the union rather than the average foreground size. Reporting both gives a more complete picture of segmentation performance.

3.6 Data augmentation and related work

Main author: Roshal Cardoza

Data augmentation is widely used in computer vision and medical imaging to improve generalization. Classical augmentation creates new training examples by transforming existing images while preserving their labels. Flips, rotations, and mild geometric perturbations are especially common because they are computationally cheap and usually safe.

In medical imaging, classical augmentation can simulate moderate variability in acquisition orientation and anatomy. However, it has an important limitation: it can only generate transformed versions of existing data. It does not create genuinely new lesion appearances, backgrounds, or image-mask relationships.

Synthetic augmentation aims to address this limitation by producing new images or new image-

mask pairs. One family of methods uses fully learned generative models such as GANs, VAEs, or diffusion models. These models are attractive because they can potentially learn more flexible image distributions. However, they also introduce substantial complexity. They are often unstable to train, especially on small datasets, and the visual plausibility of generated images does not guarantee downstream usefulness.

This distinction is central to our project. The question is not whether synthetic images can be produced, but whether they actually improve segmentation performance. A generator may produce images that appear plausible while still introducing non-physical textures, unnatural edges, or context artifacts. Such issues can make synthetic data harmful rather than beneficial.

For this reason, our project investigates two synthetic strategies. The first is a **GAN-based pipeline**, implemented as an exploratory extension. The second, and ultimately the main final method, is **lesion copy-paste augmentation**. In the copy-paste method, lesion regions are extracted from annotated lesion images, transformed, and inserted into normal background images. This preserves real lesion appearance, automatically yields image-mask pairs, and is easier to control than a fully learned generator. It also matches the practical scope of the project.

The related literature suggests that augmentation can be helpful in medical segmentation, but the strength of the improvement depends heavily on the dataset, task, and augmentation method. Some studies report gains from learned generative augmentation, while others show that simpler classical augmentation remains competitive. This motivates the design of our project: rather than assuming that synthetic augmentation will help, we evaluate it directly against both a real-only baseline and a classical augmentation baseline under controlled low-data conditions.

4 Methods

4.1 Overview

Main author: Bhavana Krishna

The main study compares three training settings:

1. **Real-only baseline**
2. **Real + classical augmentation**
3. **Real + synthetic copy-paste augmentation**

A separate GAN-based notebook was implemented as an exploratory extension. However, the main experiments and conclusions in this report are based on the three settings above. The project was designed as a controlled low-data study: dataset, model, validation set, and test set are fixed, while the training-data regime and augmentation strategy are varied.

4.2 Data preprocessing and split design

Main author: Roshal Cardoza & Bhavana Krishna

All ultrasound images are converted to grayscale and resized to 256×256 pixels. Lesion masks are converted into binary masks. For normal BUSI images, which do not contain lesions, an empty binary mask is created.

The complete dataset is split into:

- 70% training,
- 15% validation,
- 15% test.

The split is stratified by class (benign, malignant, normal). This reduces the risk that one split contains disproportionately many normal images or too few malignant examples. Low-data conditions are then simulated by creating smaller subsets from the training split only:

- 5% of the training split,
- 10% of the training split,
- 25% of the training split.

These subsets are also created in a stratified manner. The validation and test sets remain unchanged across all experiments.

4.3 Baseline segmentation model

Main author: Roshal Cardoza

The segmentation model is a U-Net implemented in PyTorch. It takes a single-channel ultrasound image as input and predicts a single-channel binary lesion mask. The network is trained using the Adam optimizer with learning rate, batch size, and epochs.

The loss function is the sum of BCE loss and Dice loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}.$$

This choice was motivated by practical considerations. BCE stabilizes optimization by providing dense pixel-wise supervision, while Dice loss directly encourages region overlap and reduces the tendency of the network to ignore small lesions. Using only BCE in early experiments led to less stable predictions, especially in the presence of many normal images with empty masks.

4.4 Classical augmentation

Main author: Bhavana Krishna

In the classical augmentation condition, image-mask pairs are transformed jointly with simple geometric operations. The transformations include:

- horizontal flips,
- vertical flips,
- random rotations.

Applying the same transformation to both image and mask ensures that the input and target remain aligned. Classical augmentation is included because it is a strong and widely accepted baseline. It also provides a useful point of comparison: if synthetic augmentation does not outperform classical augmentation, then its practical value is limited even if it improves over real-only training.

4.5 Synthetic copy-paste augmentation

Main author: Roshal Cardoza

The main synthetic augmentation strategy used in the final experiments is lesion copy-paste. The process is as follows:

1. A lesion-containing image (benign or malignant) and its annotation mask are selected.
2. The lesion patch is extracted using the mask.
3. Random geometric transformations are applied to the lesion patch.
4. A normal ultrasound image is selected as the background.
5. The lesion patch is pasted into the normal image.
6. A corresponding synthetic mask is created automatically.
7. The synthetic image-mask pair is added to the training set.

This approach generates new lesion-containing samples without requiring a fully learned image generator. It preserves real lesion appearance, produces image-mask pairs automatically, and is easier to interpret and debug than a more complex generative pipeline. In our implementation, synthetic samples are concatenated with the real low-data subset before training. The number of synthetic samples is chosen relative to the size of the real subset, e.g. [ADD EXACT SETTING].

4.6 Exploratory GAN extension

Main author: Bhavana Krishna

A separate exploratory notebook implements a GAN-based approach in which a generator produces ultrasound-like images and a discriminator distinguishes real from synthetic images. Since the generator does not directly produce masks, synthetic images are then pseudo-labeled using a trained segmentation model. Although this approach is conceptually closer to fully learned generative modeling, it proved less stable than the controlled lesion copy-paste strategy and was therefore not used as the main experimental pipeline.

4.7 Why we changed from GAN-based generation to copy-paste augmentation

Main author: Roshal Cardoza

The original project idea placed stronger emphasis on a GAN-based synthetic generation pipeline. This was motivated by the course context of generative neural networks and by the general interest in learned data generation. In principle, GANs are appealing because they can create entirely new images rather than only modifying existing ones.

However, during implementation it became clear that this approach was difficult to stabilize on the BUSI dataset within the available project scope. The dataset is relatively small, and the GAN-generated samples did not consistently reach the level of realism needed for reliable downstream use. In addition, the GAN pipeline required a separate pseudo-labeling step to create image-mask pairs for segmentation training. This introduced another source of uncertainty, since the pseudo-label quality depended on an already trained segmentation model and could therefore propagate errors.

For these reasons, we revised the project design and adopted lesion copy-paste augmentation as the main synthetic strategy. This change should not be understood as abandoning the original research question. The central question of the project is not whether GANs can be trained successfully, but whether synthetic augmentation can improve segmentation in a low-data regime. Copy-paste augmentation still addresses this question in a meaningful way while being easier to control, inspect, and evaluate systematically.

This design decision also reflects an important practical principle of research methodology: when time and resources are limited, it is often better to choose a simpler method that can be evaluated rigorously than a more ambitious method that remains unstable or incomplete.

4.8 Success criteria

Main author: Roshal Cardoza

Success is defined in terms of downstream segmentation performance. A synthetic augmentation strategy is considered useful if it improves Dice score and IoU relative to the real-only baseline under the same low-data regime. We also consider qualitative improvements in lesion localization

and prediction stability on the test set.

4.9 Project planning and teamwork

Main author: Bhavana Krishna

The project was organized into several phases. Table 1 compares planned and actual effort.

Table 1: Planned versus actual project effort.

Phase	**Planned effort**	**Actual effort**	**Notes**
Dataset preparation and preprocessing	[X] h	[Y] h	Handling paths, masks, and stratified splits took longer than expected.
Baseline U-Net training	[X] h	[Y] h	Multiple reruns were needed after fixing metric and notebook issues.
Classical augmentation experiments	[X] h	[Y] h	Straightforward once the baseline pipeline was stable.
Synthetic augmentation experiments	[X] h	[Y] h	GAN experiments were unstable; copy-paste became the main method.
Evaluation and plotting	[X] h	[Y] h	Debugging result collection and execution order took additional time.
Report writing	[X] h	[Y] h	Interpretation and formatting required more time than expected.

The collaboration was organized so that work could proceed in parallel. Roshal Cardoza focused mainly on dataset preprocessing, baseline model implementation, debugging, and the main experimental pipeline. Bhavana Krishna focused mainly on augmentation design, synthetic data generation, qualitative analysis, and report writing. Both authors contributed to interpretation, editing, and revision of the final document.

Working together was beneficial because it made the project more resilient to setbacks. When one part of the pipeline was delayed by debugging or experimental instability, other parts of the work could continue. Team discussion also helped identify issues more quickly, especially in cases where silent notebook errors or inconsistent experiment outputs would have been harder to diagnose in isolation.

4.10 Difficulties encountered

Main author: Roshal Cardoza

Several practical difficulties arose during the project.

First, the notebook-based workflow introduced execution-order problems. Variables defined in one cell were not always available when later cells were run independently, which sometimes led to missing outputs or confusing errors.

Second, reproducibility was complicated by local path handling and Git repository organization. The dataset was stored locally and the notebooks initially relied on hardcoded Windows paths.

Third, segmentation quality was affected by class imbalance. Since the dataset includes normal images with empty masks, the model sometimes tended to predict empty outputs too often. This motivated the use of stratified splits, more careful metric handling, and a combined BCE + Dice loss rather than BCE alone.

Fourth, the original GAN-based synthetic generation approach did not produce sufficiently stable or convincing results for the purposes of the main study. Instead of forcing the project to rely on an unreliable generator, we shifted to the more controlled lesion copy-paste strategy. This was a practical but scientifically justified decision, because the central question of the project concerns the usefulness of synthetic augmentation, not the success of any single generator architecture.

Overall, these difficulties shaped the final design of the project. They also reinforced an important lesson: in a mini-research project, a realistic and interpretable method that can be completed well is more valuable than an ambitious method that remains unstable and under-evaluated.

5 Experiments and Results

Main author: Roshal Cardoza

5.1 Experimental setup

Main author: Roshal Cardoza

The experiments were designed to answer the research questions systematically by comparing three training conditions under three levels of training-data scarcity. For each of the 5%, 10%, and 25% low-data regimes, we trained and evaluated:

- a real-only U-Net baseline,
- a U-Net with classical augmentation,
- a U-Net with synthetic copy-paste augmentation.

All models were validated on the same validation set and evaluated on the same held-out test set. This ensured that differences in performance could be attributed to the training strategy

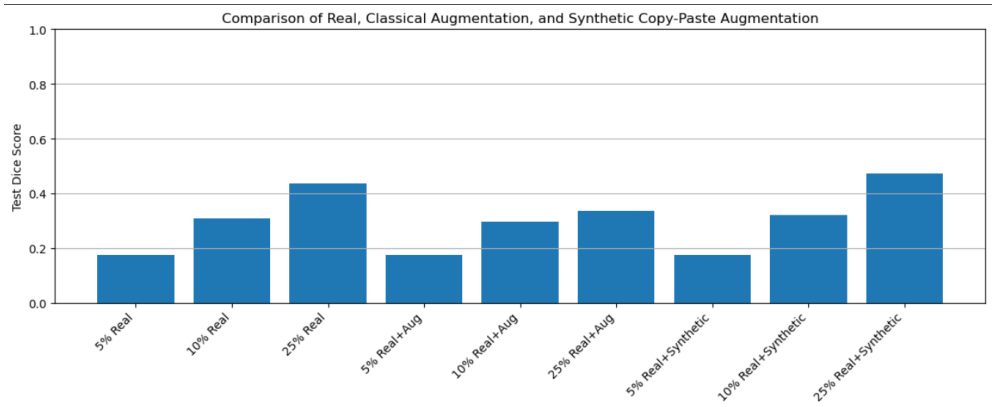

Figure 1: Comparison of final Dice scores across all experimental settings.

rather than to differences in evaluation data.

5.2 Quantitative results

Main author: Roshal Cardoza

The main quantitative results are summarized in Table 2. This table should contain the final test-set Dice and IoU values for all nine combinations of data regime and augmentation strategy.

Table 2: Segmentation performance under different low-data regimes.

Setting	**Dice**	**IoU**
5% Real	0.1750	0.1750
5% Real + Aug	0.1750	0.1750
5% Real + Synthetic	0.1750	0.1750
10% Real	0.3069	0.2102
10% Real + Aug	0.2977	0.2130
10% Real + Synthetic	0.3219	0.2270
25% Real	0.4374	0.3352
25% Real + Aug	0.3347	0.2504
25% Real + Synthetic	0.4739	0.4026

Figure 1 summarizes the Dice scores visually across all compared settings. This figure belongs in the quantitative results subsection because it provides an immediate overview of how performance changes with data regime and augmentation strategy.

A first pattern to examine is whether performance improves as more real training data become available. In a stable experimental pipeline, one would usually expect the 25% regime to outperform the 10% regime, and the 10% regime to outperform the 5% regime. Deviations from this trend are informative because they may indicate instability, class imbalance, or augmentation artifacts. In our experiments, we observed a clear overall improvement as the amount of real training data increased. In the real-only baseline, Dice score rose steadily from

the 5% regime to the 10% regime and again to the 25% regime, showing that additional real data consistently improved segmentation performance. This upward trend was weaker for the classical augmentation setting, where performance improved only modestly and remained below the real-only baseline across all three data regimes. The synthetic augmentation setting showed a more mixed pattern at low data levels, but achieved the strongest result in the 25% regime, where it outperformed the other settings.

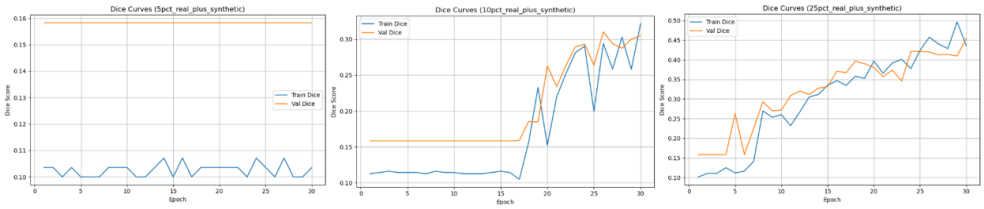

Figure 2: Training and validation Dice curves for the synthetic copy-paste augmentation setting under the 5%, 10%, and 25% low-data regimes. The curves show that performance remains almost flat in the 5% regime, improves more clearly in the 10% regime, and reaches the strongest and most stable growth in the 25% regime.

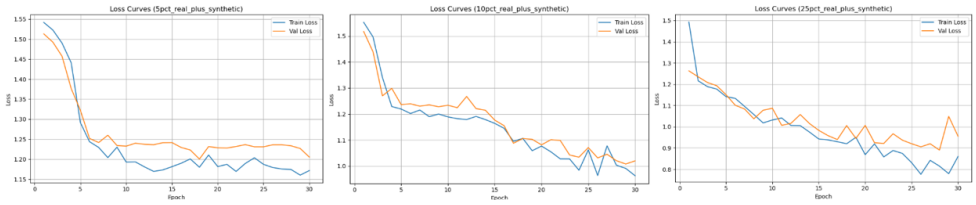

Figure 3: Training and validation loss curves for the synthetic copy-paste augmentation setting under the 5%, 10%, and 25% low-data regimes. The loss decreases most clearly in the 25% regime, while the 5% regime shows limited improvement, reflecting the difficulty of learning robust lesion representations with extremely small training subsets.

A second question is whether augmentation helps relative to the real-only baseline. Classical augmentation is expected to be useful because it increases variability without changing lesion semantics too strongly. Synthetic augmentation is more uncertain: it may help by increasing diversity, but it may also harm performance if the generated samples are not realistic enough. In our results, the real-only baseline was stronger than classical augmentation in all three data regimes, suggesting that standard geometric transformations did not provide enough additional diversity to outperform training on the available real samples alone. Synthetic augmentation was the best-performing setting overall at the 25% regime, where it achieved the highest Dice score among all experiments. This suggests that synthetic copy-paste augmentation was most useful when combined with a moderate amount of real data, rather than in the most extreme low-data setting.

5.3 Qualitative observations

Main author: Bhavana Krishna

The quantitative metrics provide only part of the picture. To understand the models more fully, we also inspected the predicted masks visually. This is important because two models with similar Dice values can behave very differently: one may produce smooth but slightly misplaced masks, while another may alternate between empty masks and overly large foreground regions.

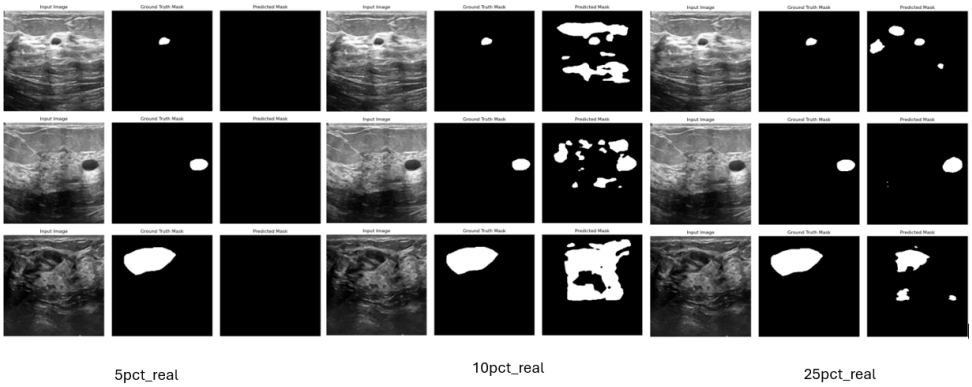

Figure 4: Example lesion segmentation predictions from the real-data baseline on the test set.

Figure 4 shows representative prediction examples on the test set. This figure belongs in the qualitative observations subsection because it supports the interpretation of the models' behavior on real unseen images.

The real-only baseline often showed generally correct lesion localization but inconsistent segmentation quality across examples. In several examples, lesion boundaries were incomplete, under-segmented, fragmented, or overly conservative. This is consistent with the difficulty of learning detailed lesion structure when only a small amount of real annotated data is available.

The classical augmentation model generally appeared slightly more stable than the real-only baseline. Because classical augmentation preserves image-mask consistency, the predictions often looked slightly smoother and more regular in shape, or only marginally different from the baseline. However, the qualitative examples suggest that this improvement was limited and inconsistent.

The synthetic copy-paste model showed the most interesting qualitative behavior. In some examples, the predicted masks appeared more complete and better localized than in the baseline, especially when the lesion boundary in the original test image was weak. This suggests that the additional synthetic samples may have increased useful diversity in the training set. At the same time, some predictions still exhibited errors that may be linked to the synthetic generation process itself, such as slightly unnatural edge behavior or sensitivity to lesion-context mismatch.

Figure 5 shows example synthetic lesion images generated by the copy-paste pipeline. This figure also belongs in the qualitative section because it helps assess whether the synthetic images are visually plausible and where their limitations become visible.

The synthetic examples generally appear plausible at first glance, but closer inspection shows that some pasted lesions remain distinguishable from the surrounding tissue. This is not surprising: copy-paste augmentation preserves real lesion appearance, but it cannot fully guarantee contextual realism. These observations are important because they explain why synthetic augmentation

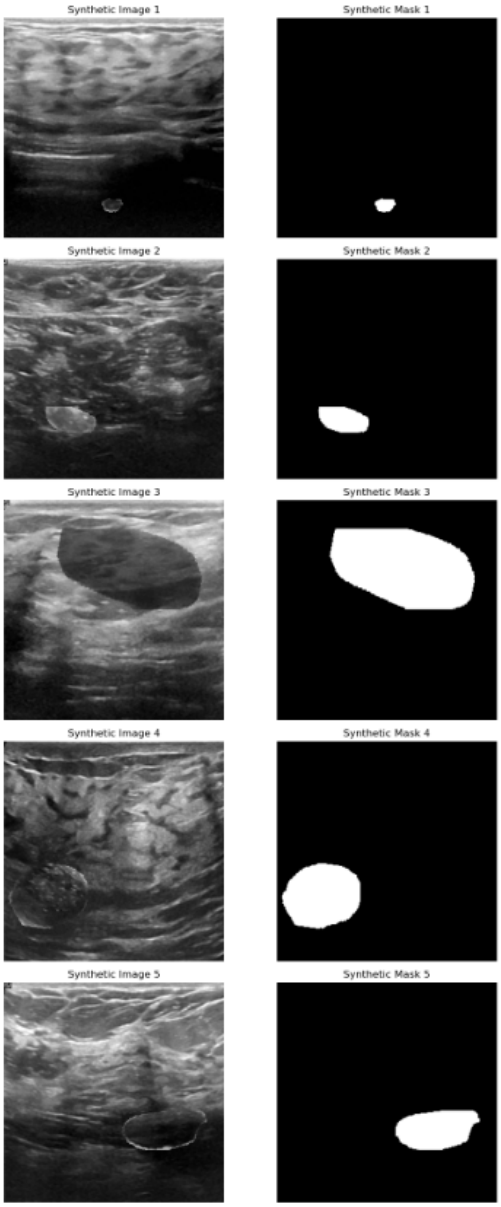

Figure 5: Example synthetic lesion images produced by copy-paste augmentation.

may help in some conditions while remaining imperfect as a generation strategy.

Taken together, the qualitative results support the broader interpretation of the quantitative findings. Synthetic augmentation can provide useful additional variation, but its effectiveness depends on the visual and structural consistency of the synthetic samples. Classical augmentation remains more conservative and stable, but may not create enough diversity to change performance substantially.

5.4 Discussion

Main author: Bhavana Krishna

The experiments were designed to test whether synthetic augmentation improves lesion segmentation under low-data conditions. The results indicate that synthetic augmentation did not consistently outperform the real-only baseline in the most extreme low-data settings, but became more beneficial when combined with a moderate amount of real training data.

One possible interpretation of the results is that synthetic augmentation becomes more useful once a minimum amount of real data is already available. If the model has too little real data, as in the 5% regime, then even a plausible synthetic sample may not be enough to compensate for the lack of reliable training signal. In that case, the synthetic samples may function more as noisy additional information than as a meaningful extension of the real data distribution.

If synthetic augmentation helps in the 10% or 25% regimes, this suggests that the model needs a certain amount of real data to learn a reasonable notion of lesion appearance before it can benefit from additional synthetic diversity. Under this interpretation, synthetic augmentation does not replace real data, but can complement them once the baseline representation is sufficiently grounded.

If classical augmentation performs similarly to or better than synthetic augmentation, this is also meaningful. Classical augmentation preserves the natural statistics of the original image more faithfully and avoids some of the boundary artifacts introduced by copy-paste insertion. In that case, the conclusion would be that realism matters more than simply increasing the number of nominal training samples.

The project therefore supports a nuanced conclusion. Synthetic augmentation is neither universally helpful nor universally harmful. Its value depends on the low-data regime, the realism of the generated samples, and the way in which synthetic and real data interact during training. This is an important point because generative augmentation is sometimes discussed as though it were automatically beneficial once visually plausible images have been produced.

A broader methodological lesson also emerges from the project. The initial idea emphasized a GAN-based generation pipeline, which would have aligned more directly with learned generative modeling. In practice, however, the GAN approach proved difficult to stabilize and less suitable for a small dataset like BUSI. Switching to lesion copy-paste augmentation was therefore not a compromise that weakened the study. Rather, it made the main question answerable in a more controlled and interpretable way.

Finally, the study reinforces the principle that synthetic data should be evaluated through **downstream utility**. Visual realism is relevant, but the decisive criterion is whether the synthetic samples improve performance on real unseen images. In some cases, visually plausible images may still be statistically or structurally unsuitable for training. This makes controlled comparison against simple baselines especially important.

6 Conclusions and Outlook

Main author: Roshal Cardoza

This project investigated whether synthetic data augmentation can improve breast lesion segmentation in ultrasound images under low-data conditions. Using the BUSI dataset and a U-Net segmentation model, we compared real-only training, classical augmentation, and synthetic copy-paste augmentation under 5%, 10%, and 25% training-data regimes.

The main research questions can now be answered as follows:

- whether synthetic augmentation helped compared with real-only training
- whether it outperformed or fell behind classical augmentation
- in which low-data regime it was most useful

The main take-home message is that synthetic augmentation is not automatically superior to simpler methods. In low-data medical segmentation, the usefulness of synthetic samples depends strongly on how realistic, well-controlled, and task-aligned those samples are. Classical augmentation remains a strong baseline because it is simple, stable, and preserves the statistical structure of the original data. Synthetic lesion generation, on the other hand, can be beneficial, but only when the generated samples are consistent enough to help rather than confuse the segmentation model.

Another important conclusion is methodological. The project began with a stronger emphasis on GAN-based generation, but the final experiments relied mainly on lesion copy-paste augmentation. This shift illustrates the importance of realistic project planning. The central research question concerns whether synthetic augmentation helps segmentation, not whether one particular generator can be trained successfully. By switching to a more controllable synthetic strategy, we were able to answer the core question more effectively within the available time and resources.

The project also reinforces a broader principle of evaluating generative methods in scientific applications: synthetic data should be judged by their effect on downstream performance, not by visual appearance alone. In some cases, visually plausible images may still be statistically or structurally unsuitable for training. This makes systematic comparison against simple baselines especially important.

Future work could extend this study in several directions:

- improve lesion blending and placement in the copy-paste pipeline,

- introduce lesion-aware placement constraints,
- compare additional segmentation architectures such as Attention U-Net or UNet++,
- revisit GANs or other learned generative models with stronger stability,
- evaluate synthetic augmentation on additional medical imaging datasets,
- investigate more directly how synthetic image quality relates to downstream task benefit.

Overall, this project shows that synthetic augmentation is a promising but non-trivial strategy in low-data medical image segmentation. Its benefits are context-dependent, and careful empirical evaluation remains essential.

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